Write legibly showing all the steps VERY clearly. Max. score=100.

1. Let m be the Lebesgue measure on $[0,1]$ and $I_A(x)$ denote the indicator function of A. Consider the following sequence of random variables on $(\Omega = [0,1], \mathcal{F} = \mathcal{B}[0,1], P = m)$: For $n \geq 1$, $X_n(\omega) = I_{\left(0, \frac{n+1}{2n}\right)}(\omega)$ and $X(\omega) = I_{\left[0, \frac{1}{2}\right]}$ for $\omega \in \Omega$.

   (i) Show $X_n \to X$ almost surely (P).
   (ii) Does $X_n \to X$ in probability ?
   (iii) For which values of $0 < p < \infty$, does $X_n \to X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$?

   ($3\times 10 =$ **30** points)

2. Let X, Y be two **independent** random variables on some probability space $(\Omega, \mathcal{F}, P)$, with marginal laws P_X, P_Y on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e. their joint law is the *product measure* of their marginal laws, i.e $P_{(X,Y)} = (P_X) \times (P_Y)$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let m is the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $m^2 \doteq m \times m$ is the Lebesgue measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let f, g be two non-negative functions in $L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$. - (a) If X and Y are independent and $E(X^2) < \infty,\ E(Y^2) < \infty$, then carefully prove

   $$Var(X+Y) = Var(X) + Var(Y).$$

   Note: Using induction, we can prove $Var(\sum_{i=1}^n X_i) = \sum_{i=1}^n Var(X_i)$, if $\{X_i\}$ independent (no need to prove)

   \- (b) Suppose $P_X \ll m, P_Y \ll m$, with $\frac{dP_X}{dm} = f$ and $\frac{dP_Y}{dm} = g$. Then prove

   $$P_{(X,Y)} \ll m^2, \quad \frac{dP_{(X,Y)}}{dm^2} = f.g \quad a.e\ (m^2)$$

   \- (c) If $\lambda \doteq P_X * P_Y$ Prove $\frac{d\lambda}{dm} = f * g$ a.e (m), f and g are as in (b).

   ($4\times 10=$ **30** points)

3. (i) In Exam 1, you showed that an $Exponential(1)$ random variable exists: In other words, you showed the existence of a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \to \mathbb{R}$, with c.d.f. F_X where $F_X(x) = \int_{-\infty}^x I_{(0,\infty)}(u) e^{-u} du,\ x \in \mathbb{R}$ (no need to show this here). For this X, compute the following Lebesgue integrals (State any measure theory results used, Riemann calculus facts can be used without proof)

   $$\mu \equiv E(X) \equiv \int_\Omega X(\omega) dP(\omega), \quad \sigma^2 \equiv Var(X) \equiv \int_\Omega [X(\omega) - \mu]^2 \, dP(\omega).$$

   (ii) Show that there exists $\{X_n : n \geq 1\}$ which is an i.i.d. sequence of $Exponential(1)$ random variables on some probability space $(\Omega, \mathcal{F}, P)$.

   (iii) Define $Y_n = \frac{1}{n} \sum_{i=1}^n X_i$ for $n \geq 1$. Argue that Y_n is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$-measurable. Using any of the previous questions or otherwise, compute $E(Y_n)$ and $Var(Y_n)$.

   (iv) Using any of the previous questions (and any inequalities) or otherwise, prove $Y_n \to Y$ in probability, where $Y \equiv 1$.

   ($4\times 10=$ **40** points)

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A),\ A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^{1} p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1$, $n \in \mathbb{N}$	$\sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A)$, $A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b])/(b-a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx$, $a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha,\beta)} \int_A x^{\alpha-1}(1-x)^{\beta-1} I_{(0,1)}(x) m(dx)$, where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, $a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2}\ m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x - \gamma)^2/2\sigma^2}}{x}\ I_{(0,\infty)}(x) m(dx)$